

Supplementary material

Population genomics of the northern shortfin squid *Illex illecebrosus*

September 22, 2026

Supplementary figures

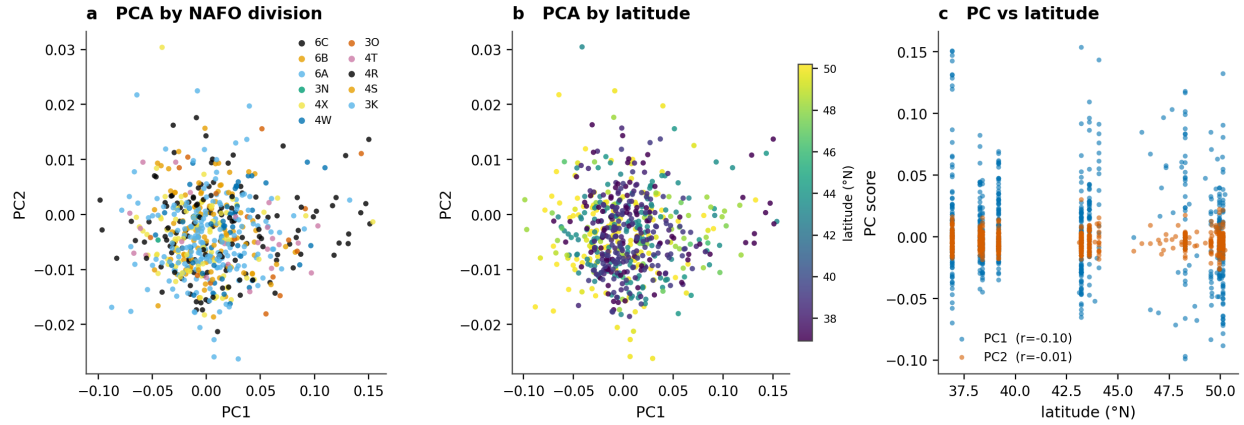


Figure S1: **The autosomal-PCA spread is not geographic structure.** Genome-wide PCA (autosomes; chromosome 2, chromosome 42 and the Z excluded), $n = 621$ individuals. (a) Coloured by NAFO sampling division and (b) by latitude, the divisions and the latitudinal gradient are fully intermixed across PC space — no clustering by origin. (c) PC scores plotted against latitude show no relationship (Pearson $r = -0.10$ for PC1, $r = -0.01$ for PC2). The dispersion along the minor PC axis in main-text Fig. 2 is carried by a handful of individuals (~ 9 samples account for $>80\%$ of the PC2 variance) and tracks neither division nor latitude nor sequencing coverage; a KING-robust screen finds no cryptic relatedness among them, so it is a minor-axis artifact rather than population subdivision, consistent with genome-wide $F_{ST} \approx 0$.

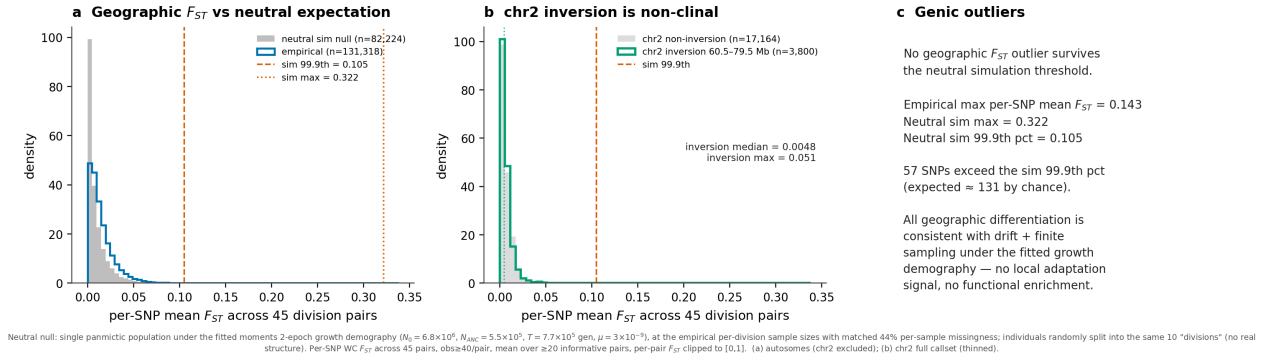


Figure S2: No geographic differentiation beyond genetic drift; no local adaptation signal. Per-SNP Weir–Cockerham F_{ST} across all 45 NAFO-division pairs (well-genotyped sites, ≥ 40 observed haplotypes per pair; per-SNP mean over ≥ 20 informative pairs). **(a)** The empirical distribution (blue) lies entirely within a coalescent *neutral null* (grey): a single panmictic population simulated under the fitted growth demography ($N_0 = 6.8 \times 10^6$, $N_{ANC} = 5.5 \times 10^5$, $\mu = 3 \times 10^{-9}$) at the empirical per-division sample sizes with the matched 44% missingness, split at random into ten “divisions”. Empirical max $F_{ST} = 0.143$ vs neutral max 0.322; only 57 SNPs exceed the neutral 99.9th percentile (0.105) against ≈ 131 expected by chance — *fewer* than the false-positive rate. Drift, finite sampling and missingness fully explain the observed differentiation. **(b)** On chromosome 2, the inverted region (green) is geographically indistinguishable from the collinear remainder (median $F_{ST} \approx 0.005$) — the inversion is non-clinal; its $F_{ST} \approx 0.37$ is between *karyotypes*, not geography. **(c)** No SNP survives the neutral threshold, so a scan for local adaptation that the diversity-based sweep tests could miss returns no candidates. (The autosomal panel samples a fixed 400 kb window per chromosome; the distribution/threshold conclusion is robust, but a genome-wide genic-outlier scan would require per-chromosome F_{ST} on the full callsets.)

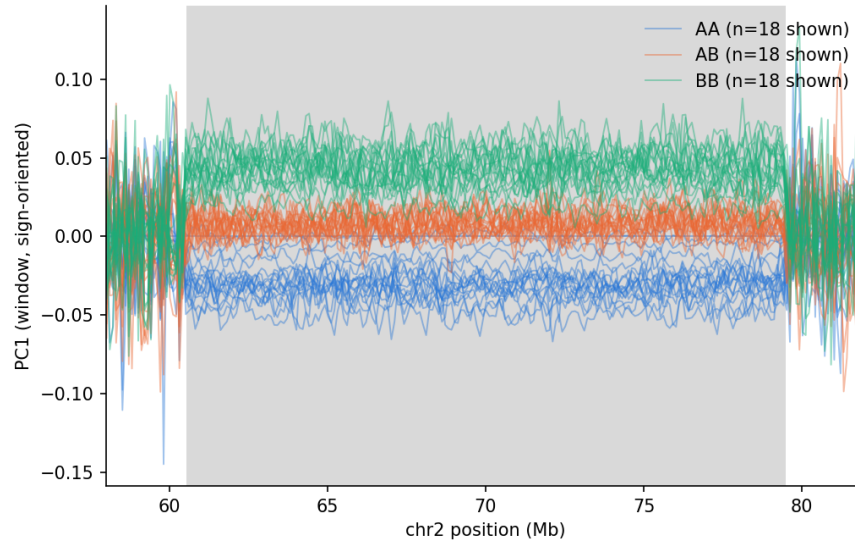


Figure S3: Windowed PCA across chromosome 2 localising the differentiated block. [Regeneration note: window coordinates to be corrected in the final version.]

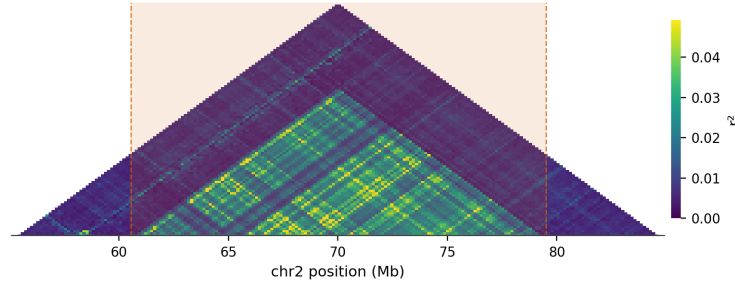


Figure S4: **Linkage-disequilibrium triangle across chr2**, showing the extended LD block coincident with the inversion.

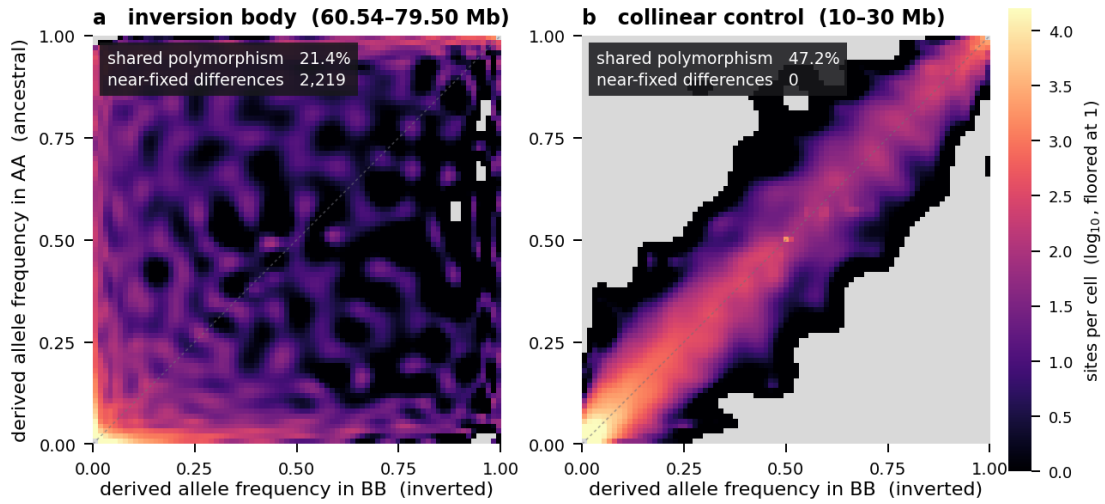


Figure S5: **Joint two-dimensional site-frequency spectrum** of derived-allele frequency in the AA (ancestral) versus BB (inverted) arrangement. Each cell is a count of sites (colour, $\log_{10}$). In the inversion body (a) density spreads off the diagonal and into the corners — variants common in one arrangement and absent in the other, i.e. near-fixed differences (2,219 sites; 21% shared polymorphism) — the direct signature of the recombination barrier. In the collinear control (b) the density collapses onto the diagonal (0 near-fixed differences; 47% shared), as expected for freely recombining, exchangeable sequence. *[Regeneration note: drop interpolation, floor counts at 1, mask empty cells to a distinct colour.]*

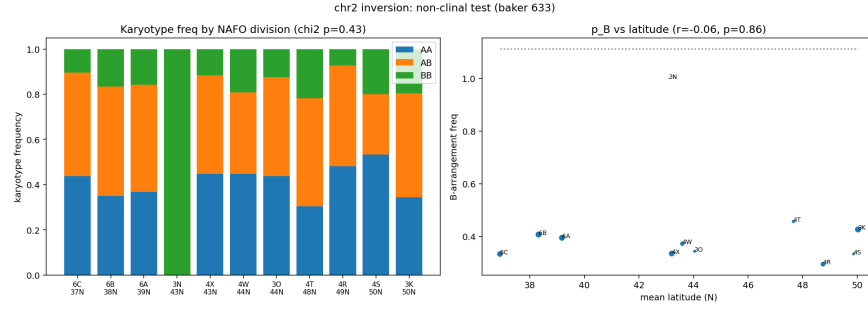


Figure S6: **The inversion is non-clinal.** Inverted-arrangement frequency across NAFO sampling divisions shows no latitudinal trend (see table S2); geographic $F_{ST} \approx 0$, and the three karyotypes co-occur throughout the range.

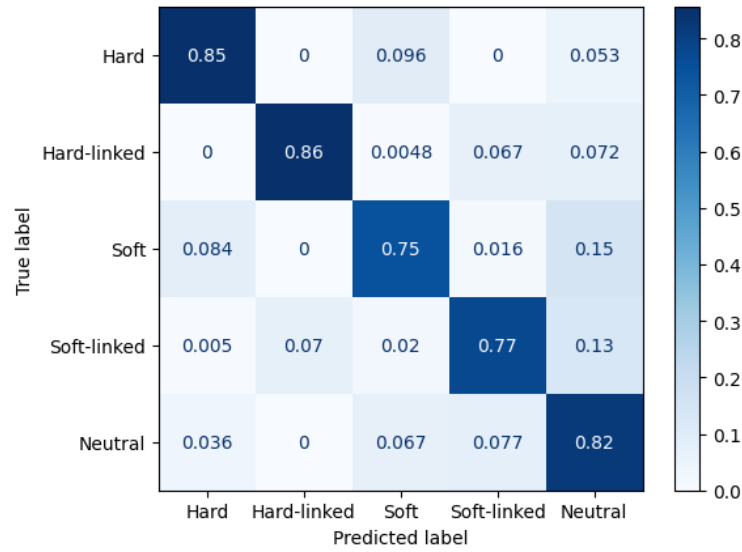


Figure S7: **Sweep-classifier confusion matrix** (five-class diploSHIC, full-SFS model; 81% overall accuracy). Recall: hard 0.85, linked-hard 0.86, neutral 0.82, soft 0.75, linked-soft 0.77.

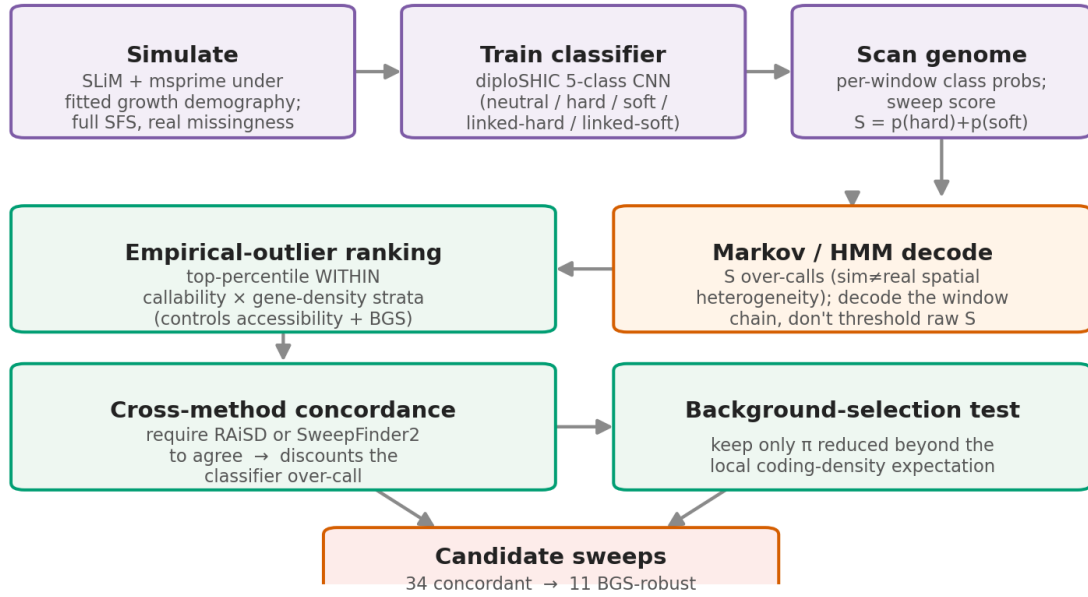


Figure S8: **The selective-sweep scan pipeline.** Simulate (SLiM + msprime under the fitted growth demography) → diploSHIC five-class classifier → Markov/HMM decode → stratified empirical-outlier ranking (within callability × gene-density strata) + cross-method concordance (RAiSD, SweepFinder2) + background-selection filter → candidate sweeps.

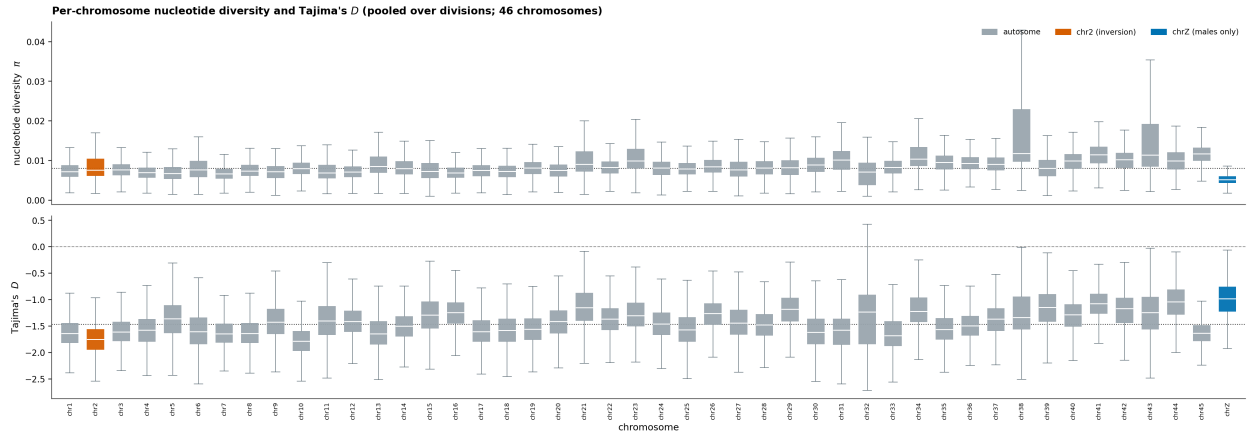


Figure S9: **Per-chromosome diversity.** Distribution of nucleotide diversity π (top) and Tajima's D (bottom) across 50 kb windows, one box per chromosome (pooled over the ten NAFO divisions; all 46 chromosomes: 45 autosomes plus chrZ). Diversity is uniform and D is strongly negative genome-wide, the signature of expansion. Two chromosomes stand apart: **chr2** (the inversion; orange) carries an inflated upper tail of π and more-negative D from the segregating arrangements, and **chrZ** (blue; computed in the 330 hemizygous-informative males only, as females are Z-haplloid) has reduced diversity (π median 0.005 vs ~ 0.008 autosomal, $\approx 0.63\times$) as expected for a sex chromosome.

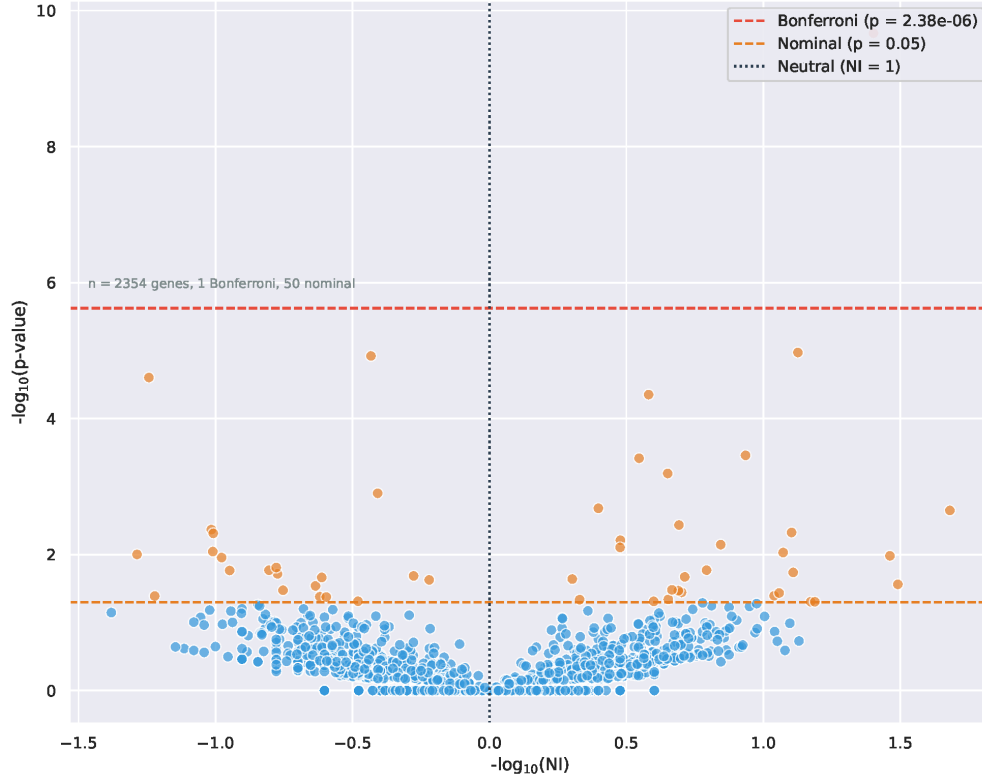


Figure S10: **Per-gene McDonald–Kreitman neutrality index** (mkado volcano plot, *I. coindetii* outgroup). Each point is one gene: $-\log_{10}$ of its neutrality index $NI = (P_n/P_s)/(D_n/D_s)$ on the x -axis (right of the neutral $NI = 1$ line = adaptive, $NI < 1$; left = excess non-synonymous polymorphism, $NI > 1$) against $-\log_{10} p$ on the y -axis. Dashed lines mark the nominal ($p = 0.05$) and Bonferroni thresholds. Of $n = 2,354$ testable genes, 50 are nominally significant and 1 survives Bonferroni; the genome-wide asymptotic $\alpha = 0.64$ resolves to few individual genes, as expected for a large- N_e species where per-gene MK is underpowered.

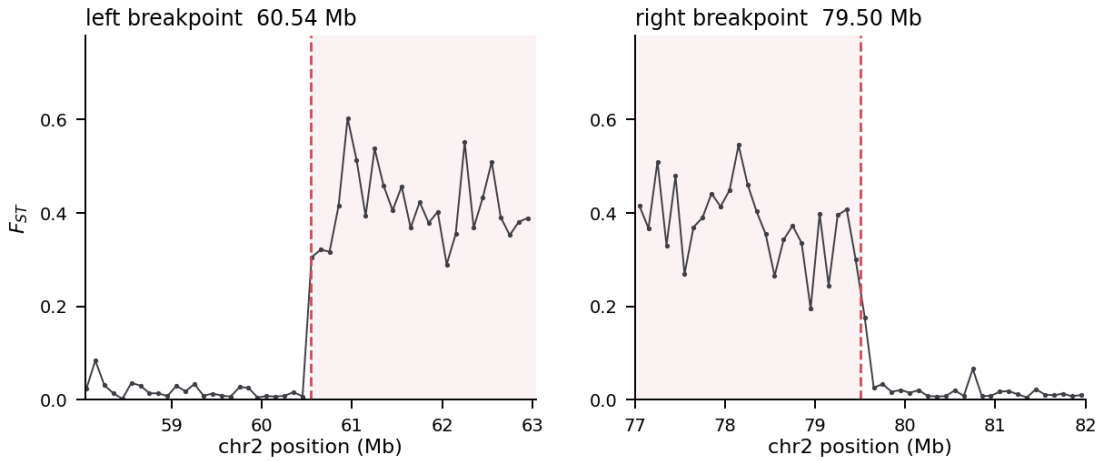


Figure S11: **Inversion breakpoints.** F_{ST} across the left (60.54 Mb) and right (79.50 Mb) breakpoints — a step, not a ramp.

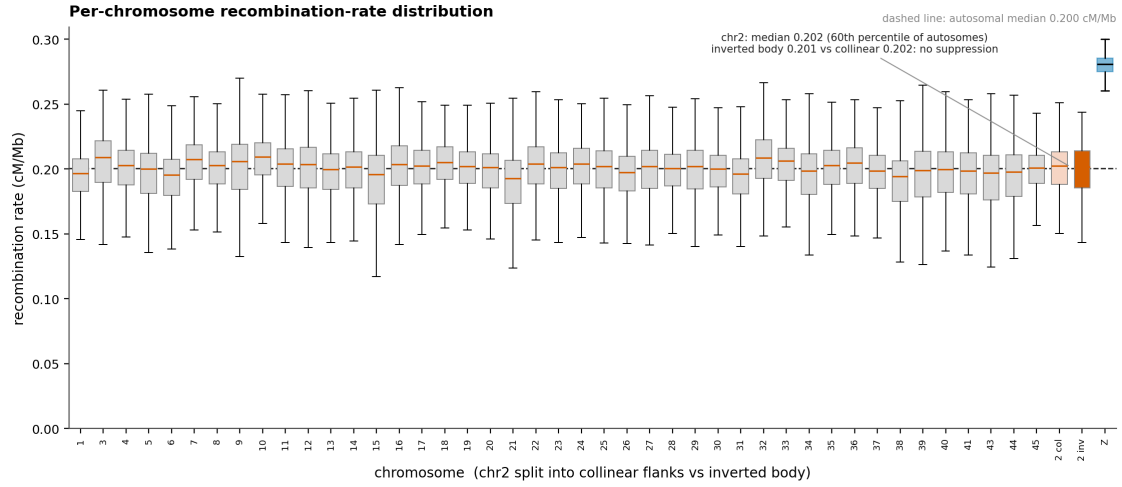


Figure S12: **Per-chromosome recombination-rate distribution** (ReLERNN). All boxes are raw ReLERNN predictions from a single network so that chromosomes are compared on the same footing: the 43 scanned autosomes, chromosome 2 split into its collinear flanks and the inverted body (60.54–79.50 Mb), and the Z. Chromosome 2 is an ordinary autosome (median 0.202 vs 0.200 cM/Mb genome-wide, the 60th percentile of per-chromosome medians) and the inverted body is not depressed relative to the flanks (0.2015 vs 0.2023): the estimator reads within-window linkage disequilibrium at the ~ 19 -kb scale and does not register the arrangement-level barrier, which is shown directly by main-text Fig. 3b and Supplementary Figs. S4 and S5.

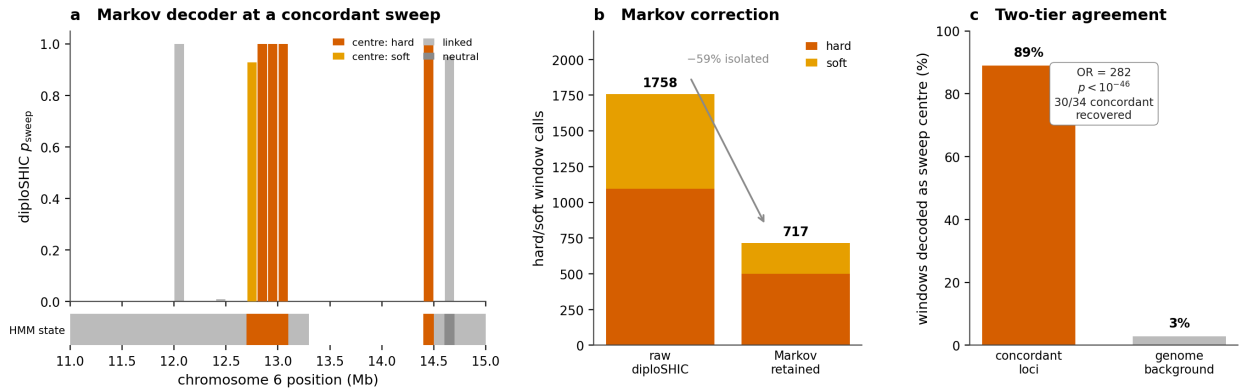


Figure S13: **Recombination-aware Markov decoder for the Tier-1 sweep calls**. A hidden Markov model over the per-100 kb-window diploSHIC classification (emissions = prior-corrected class posteriors; dwell lengths from the ~ 400 kb sweep footprint scaled by the ReLERNN map; design-informed, not simulation-calibrated). (a) The decoder at a representative concordant sweep (chromosome 6): the sweep posterior $p_{\text{hard}} + p_{\text{soft}}$ per window and the Viterbi state track (neutral / linked / centre); it localises the centre inside the linked envelope and overrides spatially isolated high-posterior windows (e.g. near 12.05 Mb). (b) Markov correction reduces the raw hard/soft calls from 1,758 to 717 spatially-coherent calls (-59%). (c) 89% of windows at the Tier-2 cross-method-concordant loci are decoded as sweep centres versus 2.9% genome-wide (odds ratio 282, $p < 10^{-46}$; 30 of 34 concordant regions recovered).

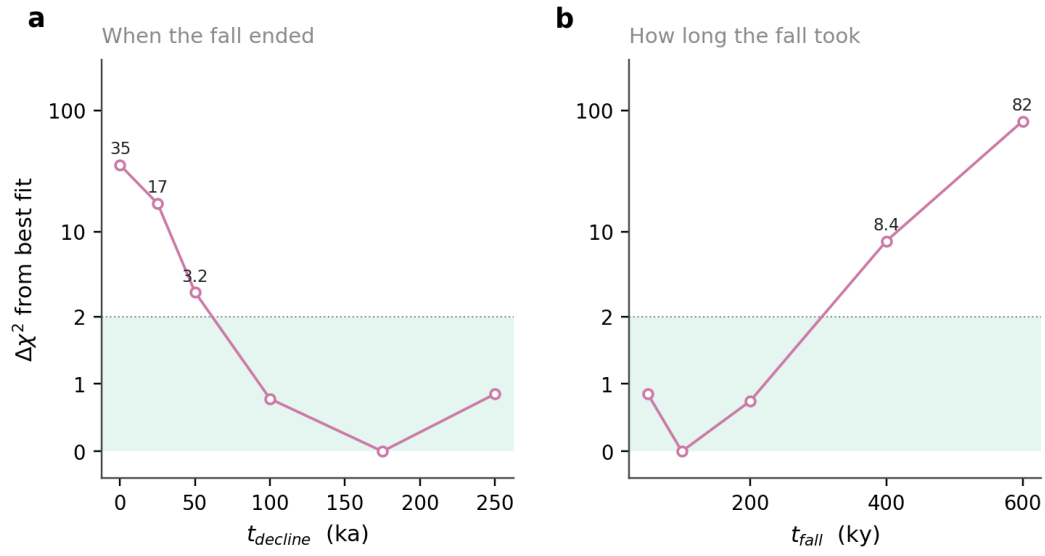


Figure S14: **Profile-likelihood bounds on the chromosome-2 inversion's decline** (main-text Fig. 4). $\Delta\chi^2$ from the best fit while profiling (a) t_{decline} , when the fall ended, and (b) t_{fall} , how long it took; the shaded band is $\Delta\chi^2 < 2$ (not excluded). The fall ended $\sim 50\text{--}200$ ka and took $\sim 50\text{--}200$ ky.

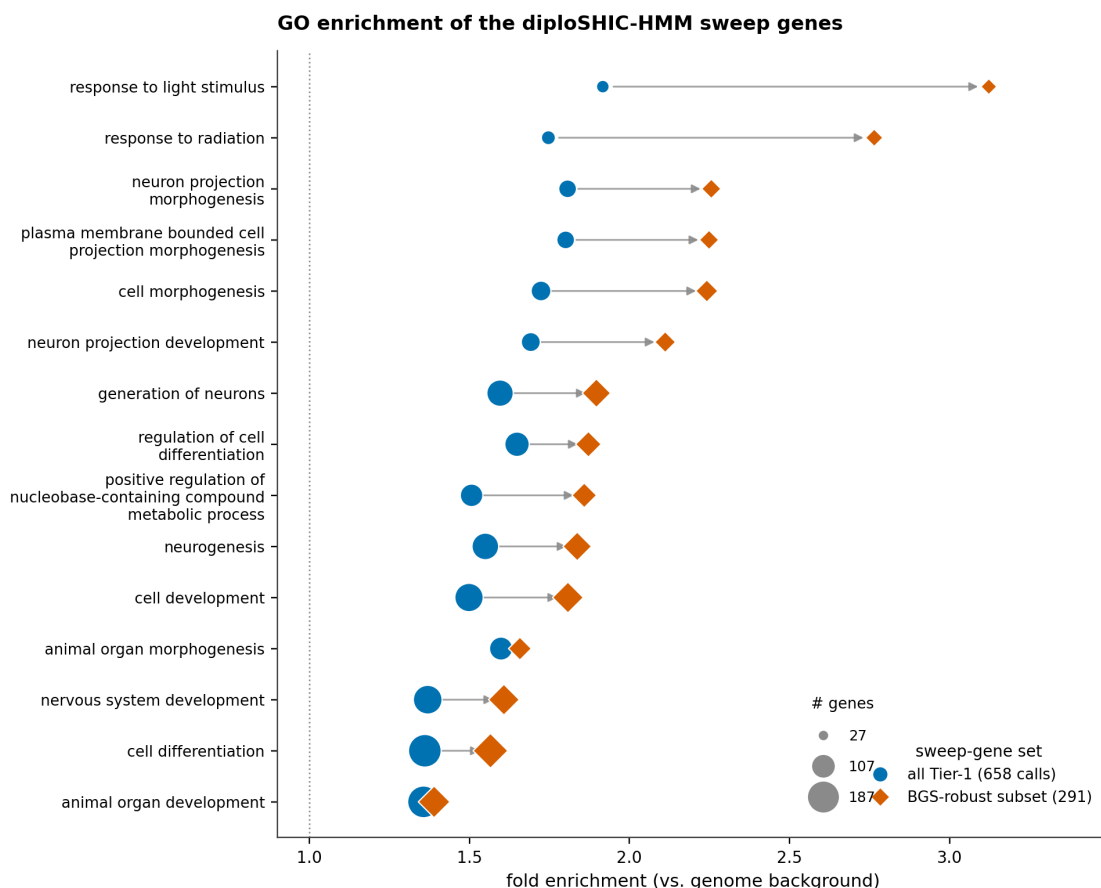


Figure S15: **GO enrichment of the diploSHIC-HMM sweep genes** (biological process; terms significant at $FDR < 0.05$ in both sets). Paired dot plot: each term is shown for the full Tier-1 set (blue circle; 658 calls) and for its background-selection-robust subset (orange diamond; 291 calls), with x = fold enrichment and marker area = number of sweep genes carrying the term. The grey arrow marks the shift between the two sets: the BGS screen consistently *sharpens* the enrichment (rightward), and most strongly for the light-response / photoreception terms (response to light stimulus $1.9 \rightarrow 3.1\times$; response to radiation $1.8 \rightarrow 2.8\times$). Two coherent themes dominate — neural development and photoreception — and both survive the background-selection filter (table S4; full per-set enrichment tables are provided with the source). The hard-sweep calls are a sub-classification recorded in the calls table, not a separate gene set.

BGS-robust sweep genes: GO enrichment map (shared-gene network)

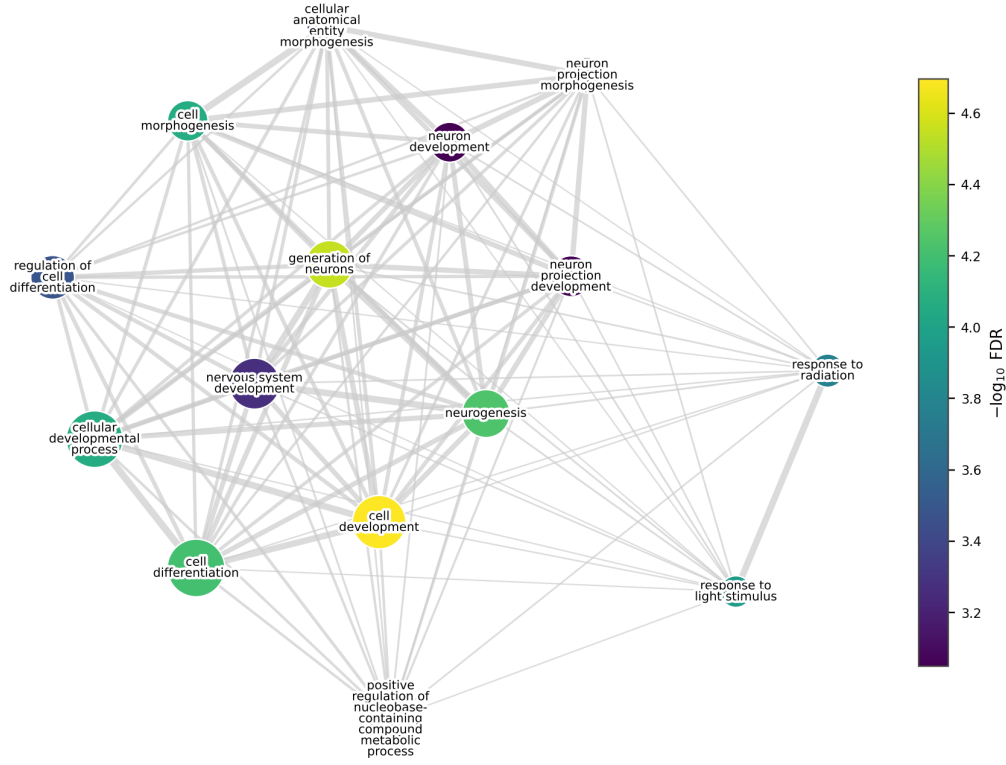


Figure S16: **GO enrichment map for the background-selection-robust sweep genes.** Built from the 291 BGS-robust Tier-1 calls (the subset whose local diversity reduction exceeds the background-selection expectation; 227 GO-annotated genes). Nodes are enriched biological-process terms (size = number of genes, colour = $-\log_{10}$ FDR); edges connect terms sharing genes (Jaccard ≥ 0.15). The terms form connected modules of neural development / cell morphogenesis and photoreception (response to light/radiation), the same two themes seen in the GSEA dot plot (fig. S15).

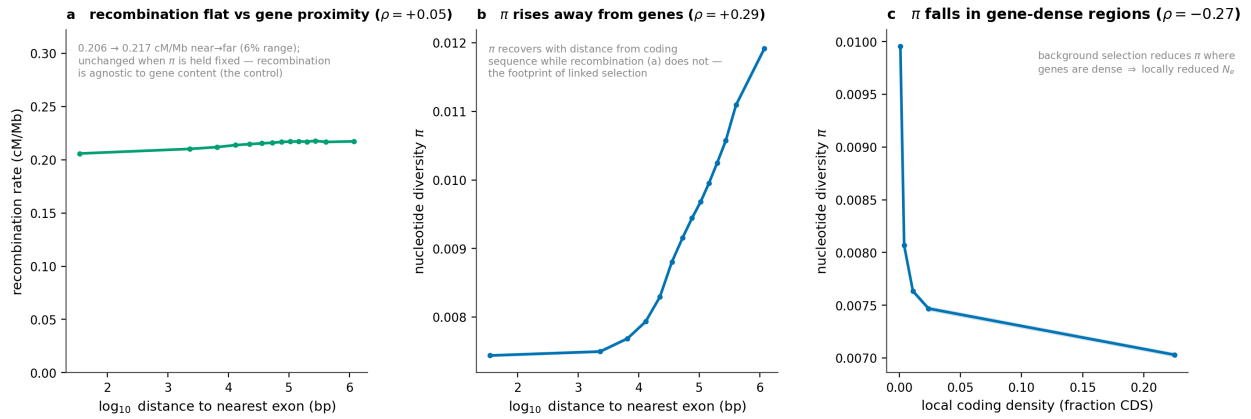


Figure S17: **Recombination, gene proximity and the efficacy of selection.** Per-window statistics (~ 50 kb windows, autosomes). **(a)** The control: recombination rate is essentially flat across distance from the nearest exon (0.206 $\rightarrow$ 0.217 cM/Mb near $\rightarrow$ far, a $\sim 6\%$ range; Spearman $\rho = +0.05$, unchanged when π is held fixed, partial $\rho = +0.06$) — recombination is agnostic to gene content, as it should be. **(b)** Nucleotide diversity π , by contrast, rises steeply with distance from coding sequence ($\rho = +0.29$; $+43\%$ near $\rightarrow$ far): the footprint of linked selection. **(c)** π falls with local coding density ($\rho = -0.27$) — background selection reduces diversity where genes are dense, i.e. a locally reduced effective size, and it is *gene-density-driven*. Linked selection in *Illex illecebrosus* is thus structured by gene density and gene proximity rather than by the recombination-rate axis, which is nearly uniform genome-wide (0.2–0.24 cM/Mb; Supplementary Fig. S22).

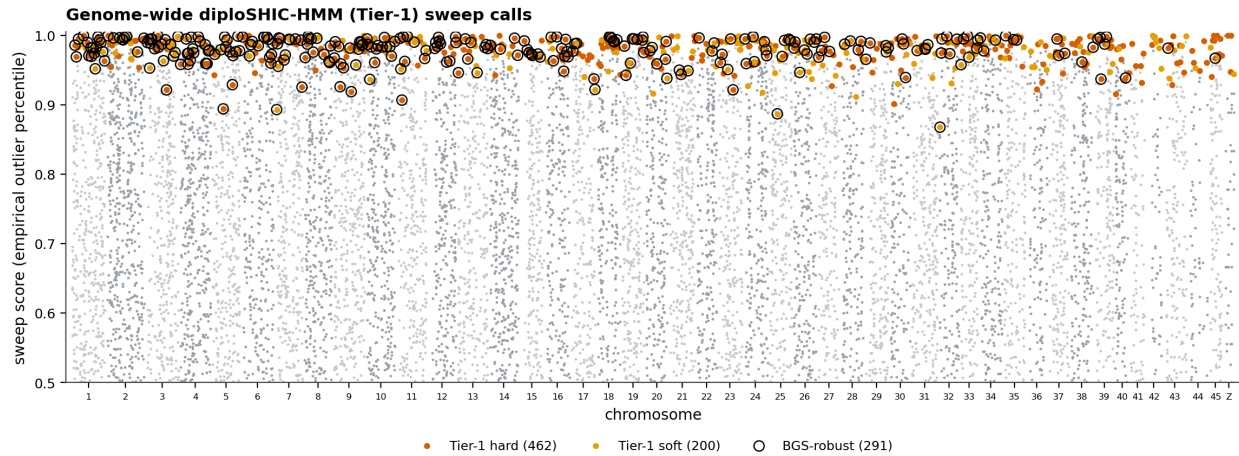


Figure S18: **Genome-wide diploSHIC-HMM (Tier-1) sweep calls.** Per-100 kb-window empirical-outlier sweep score (rank within callability $\times$ coding-density strata) across all 45 autosomes plus the Z chromosome (662 calls total; 462 hard, 200 soft). The Markov-decoded Tier-1 calls are highlighted by class (hard, soft), and the background-selection-robust subset (§ BGS screen; 291 of the 658 autosomal calls) is ringed. The four chromosome-Z calls come from the male-only ($n=330$) diploSHIC scan and are diploSHIC-only (RAiSD/SweepFinder2 do not cover the Z). The calls are distributed across the genome with no single dominant locus.

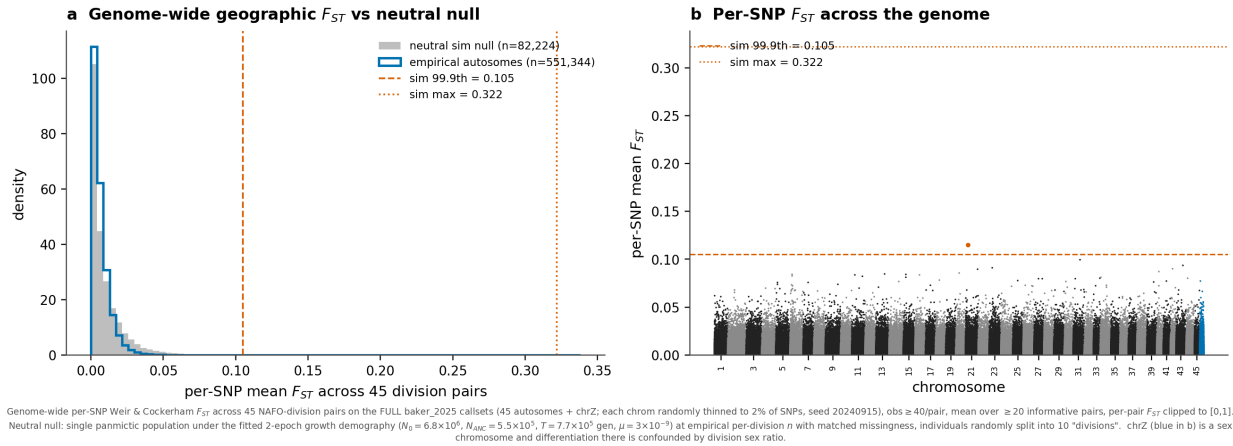


Figure S19: **Genome-wide per-SNP F_{ST} finds no differentiation outlier.** The window scan of fig. S2 extended to the *full* per-chromosome callsets (551,344 autosomal SNPs, 2% thinning, covering 12,007 genes). **(a)** The empirical per-SNP F_{ST} distribution across the 45 division pairs lies entirely within — and slightly narrower than — the coalescent neutral null. **(b)** Genome-wide, *no* SNP exceeds the neutral maximum (0.322); a single intergenic SNP on chromosome 21 exceeds the lenient 99.9th percentile (0.105), against ≈ 556 expected by chance — i.e. $\sim 555 \times$ *fewer* outliers than the false-positive rate. Zero genic outliers, hence no functional enrichment. Extreme panmixia is confirmed at genome scale, and no local-adaptation signal is present that the diversity-based sweep scans could have missed. (chrZ, blue in (b), is confounded by division sex ratios and shown for completeness.)

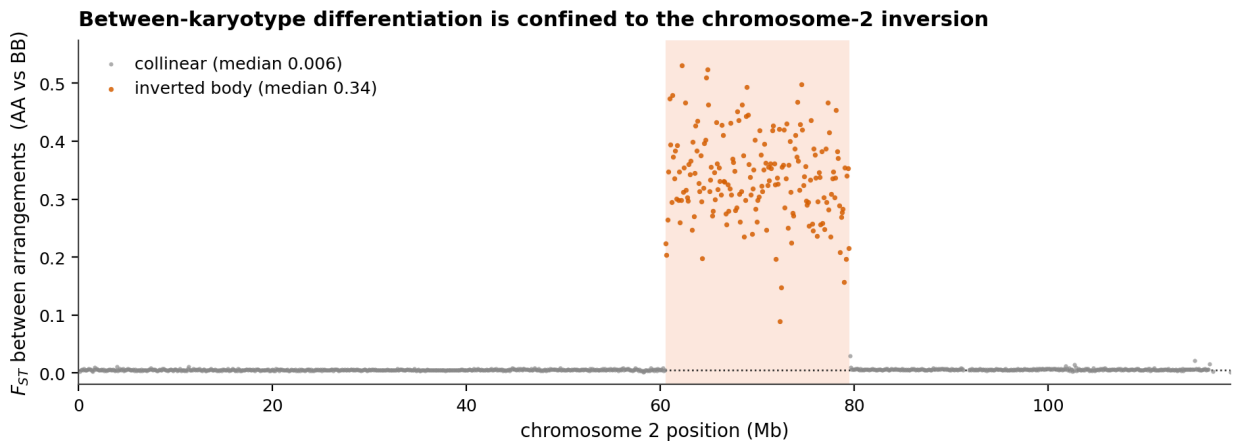


Figure S20: **Between-karyotype differentiation is confined to the chromosome-2 inversion.** Per-window F_{ST} between the AA and BB arrangements (arrangement homozygotes: AA $n = 254$, BB $n = 95$) across the whole of chromosome 2, aggregated in 100 kb windows (ratio-of-averages Hudson estimator). Unlike the population- F_{ST} scan (which asks about geography), this asks how differentiated the two arrangements are along the chromosome. The collinear remainder (grey) sits at $F_{ST} \approx 0$ (median 0.006) — outside the inversion the karyotypes are one panmictic population — while the inverted body (60.54–79.50 Mb; shaded) is strongly differentiated (median 0.34, max 0.53). The $F_{ST} \approx 0.37$ “structure” on chromosome 2 is thus between *karyotypes* and confined to the inversion, not geographic.

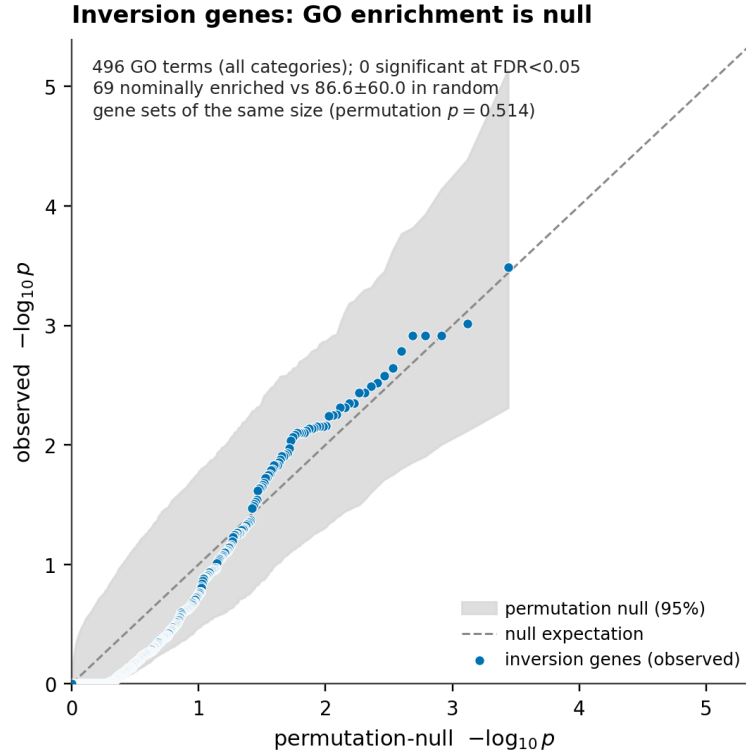


Figure S21: **The inversion-captured genes show no GO enrichment.** QQ-plot of the per-term GO-enrichment p -values (Fisher's exact vs. genome background, all three ontologies) for the genes in the inversion body against a *permutation* null (500 random gene sets of the same size). A theoretical-uniform null is inappropriate for GO — nested/correlated terms and tiny discrete counts inflate any small gene set — so the permutation band is the correct reference. The observed points fall entirely within it: the inversion genes yield *fewer* nominally enriched terms than random sets of the same size (69 vs 86.6 ± 60 ; permutation $p = 0.51$) and none survive $FDR < 0.05$. The large per-term fold values (up to $\sim 40\times$) are small-count artefacts (most rest on 2 of the 29 GO-annotated genes), not enrichment (table S7).

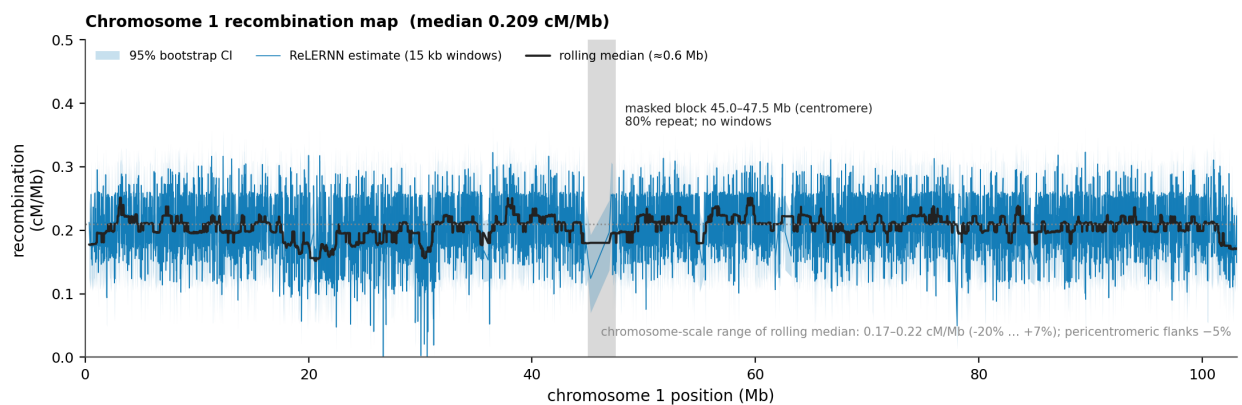


Figure S22: **The chromosome-1 recombination map, and what the LD-based map does not resolve.** ReLERNN estimate in 15 kb windows with 95% bootstrap confidence intervals and a 0.6-Mb rolling median (median 0.209 cM/Mb). The shaded 2.5-Mb block at 45.0–47.5 Mb is 80% repeat sequence (vs 52% chromosome-wide), is almost entirely masked, and carries no ReLERNN windows: the centromere. In the accessible pericentromeric flanks the estimate dips only $\sim 5\%$, although nucleotide diversity halves there (0.0038 vs 0.0084), and across the whole 103 Mb the rolling median spans just 0.17–0.23 cM/Mb (-20% to $+7\%$ of the median). The map’s chromosome-scale dynamic range is therefore compressed, so centromeric suppression cannot be resolved and the nearly uniform landscape is in part an estimator ceiling.

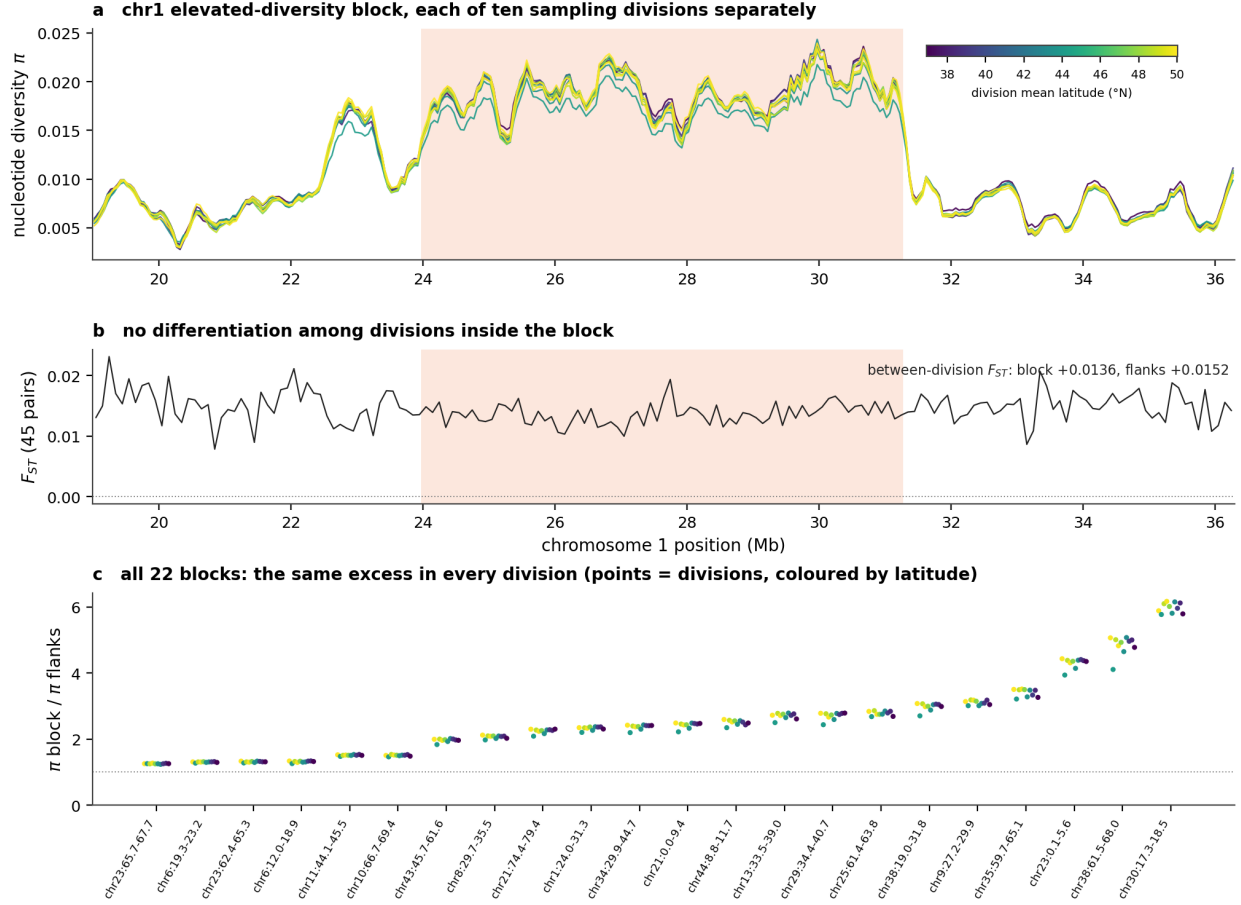


Figure S23: The 22 elevated-diversity blocks are shared by every sampling division. (a) Nucleotide diversity across chr1:19–36 Mb computed separately in each of the ten divisions with more than one individual (ANGSD, 50-kb windows, lines coloured by division latitude); the 24.0–31.3 Mb block (shaded) carries 2.2–2.4 times the flanking diversity in every division. (b) Between-division Hudson F_{ST} over the same span (all 45 division pairs, 100-kb windows) is lower inside the block than in the flanks (0.014 vs 0.015); the same holds for the three strongest blocks (chr30:17.3–18.5, chr23:0.1–5.6 and chr38:61.5–68.0 Mb; 0.006–0.007 inside vs 0.014–0.015 outside). (c) Block-to-flank diversity ratio per division for all 22 blocks: the excess varies by only 1–6% among divisions, so no block is driven by a subset of the sample. The blocks are gene-poor (a third or less of their chromosome’s gene density), largely unalignable to *I. coindetii* (chr1 block 35% vs 87–89% of flanking bases) and more divergent where aligned (3.3% vs 1.9–2.1%). On chr1 there is no depth anomaly (per-individual block-to-flank read ratio median 0.95, IQR 0.94–0.97), no excess heterozygosity, and no loss to the excess-heterozygosity filter. Across blocks, however, the diversity excess scales with a shift of Tajima’s D toward zero (Spearman $\rho = 0.75$, $p = 5 \times 10^{-5}$; strongest blocks $D \approx -0.7$ to -1.1 against ≈ -2 genome-wide), which indicates an excess of intermediate-frequency variants in the strongest blocks (Supplementary Table S9).

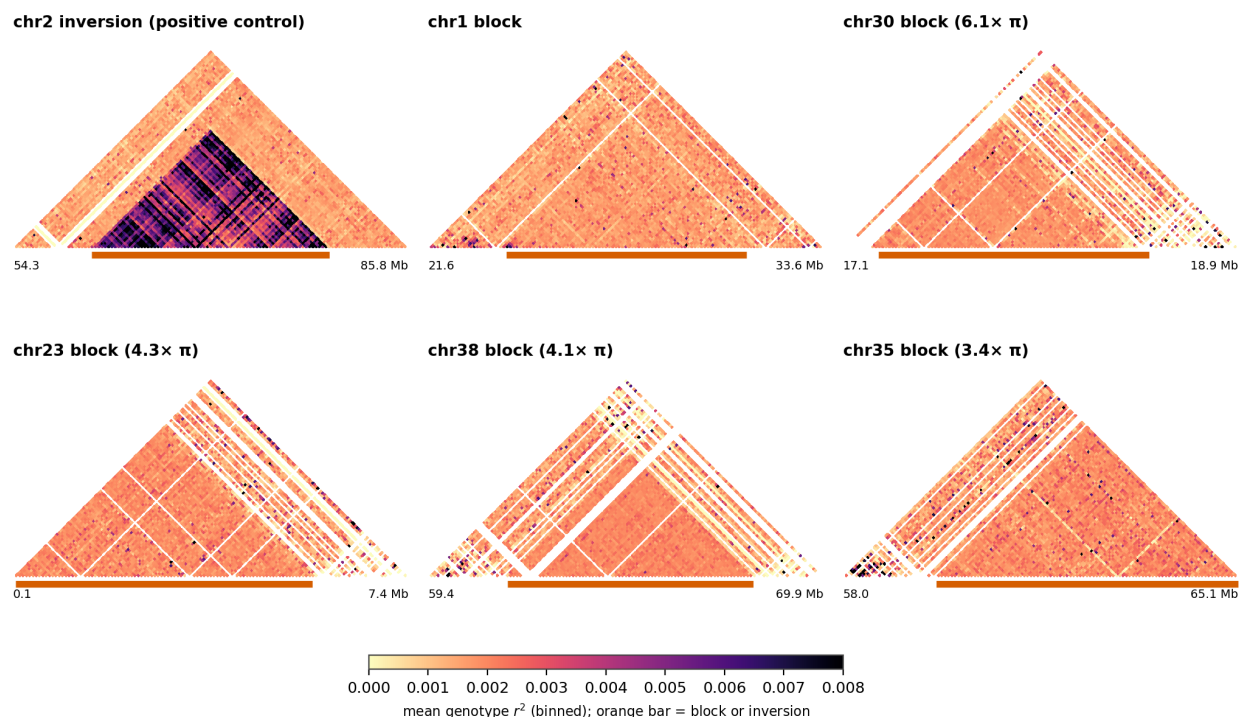


Figure S24: **The elevated-diversity blocks carry no inversion-like linkage disequilibrium.** Genotype r^2 among common SNPs (minor-allele frequency ≥ 0.1 , $\leq 30\%$ missing; 1,500 SNPs per region), averaged in 120 bins, for the chromosome-2 inversion (positive control) and five of the strongest blocks, each shown with a third of its length of flanking sequence; the orange bar marks the inversion or block. The inversion is a solid block of long-range LD over exactly its span (mean r^2 at >1 Mb 0.0056 inside vs 0.0016 in its flanks). None of the blocks differs from its flanks (0.0017–0.0020 vs 0.0012–0.0021 at >1 Mb, across all eight blocks tested), so none behaves as a segregating inversion or other long non-recombining haplotype. White stripes are bins with too few sampled SNPs.

Table S1: **Software and versions.** Versions were read off the analysis machine (conda metadata, `pip list`, `--version`, or recorded `versions.txt`). Citations are pointers to confirm at final formatting.

Tool (version)	Used for	Reference
msinv 0.3.5	chr2 inversion age/decline (inversion)	(Small, 2025); (Kelleher et al., 2016)
ANGSD 0.940	SFS, θ , diversity (demography, diversity)	(Korneliussen et al., 2014)
Stairway Plot 2 (2.1.1)	demography	(Liu and Fu, 2020)
GONE2 1.0.4	recent demography	(Santiago et al., 2025)
moments 1.6.0	demographic inference (demography)	(Jouganous et al., 2017)
demes 0.2.3	demography interchange	(Gower et al., 2022)
ReLERNN 0.2	recombination map	(Adrion et al., 2020)
diploSHIC 1.1.0	sweep scan	(Kern and Schrider, 2018)
RAiSD (0c705a8)	sweep corroboration (sweep scan)	(Alachiotis and Pavlidis, 2018)
SLiM 5.2	forward simulation	(Haller et al., 2026)
msprime 1.4.2	coalescent sim, mutation overlay	(Baumdicker et al., 2022)
tskit 1.0.3	tree-sequence I/O	(Kelleher et al., 2018)
discoal 0.1.7a	sweep-sim validation	(Kern and Schrider, 2016)
pg_gpu 0.1.1.dev14	windowed $\pi/d_{xy}/F_{ST}$, PCA	(kr-colab, 2025)
mkado 0.5.0	McDonald–Kreitman	(Rivera-Colón et al., 2026)
DIAMOND 2.1.12	homology (inversion, sweep scan)	(Buchfink et al., 2021)
HMMER 3.4	Pfam domain screen	(Eddy, 2011)
Pfam-A	domain screen	(Mistry et al., 2021)
AnchorWave 1.2.3	<i>I. coindetii</i> ↔ <i>Illex illecebrosus</i> alignment	(Song et al., 2022)
minimap2 2.29	alignment	(Li, 2018)
est-sfs 2.04	ancestral polarisation	(Keightley and Jackson, 2018)
degenotate 1.3	0-/4-fold degeneracy	(gwct, 2024)
bedtools 2.30.0	interval operations	(Quinlan and Hall, 2010)
bcftools/samtools 1.21	VCF/BAM handling	(Danecek et al., 2021)
EarlGrey 7.0.1	TE annotation	(Baril et al., 2024)
StringTie 3.0.0	transcript assembly	(Kovaka et al., 2019)
SQANTI3	Iso-Seq QC	(Pardo-Palacios et al., 2024)

Supplementary tables

Table S2: **Chromosome-2 karyotype counts and inverted-arrangement frequency by NAFO division** (ordered by latitude). The inverted (BB) frequency p_{inv} shows no latitudinal cline (range 0.29–0.46), consistent with geographic $F_{\text{ST}} \approx 0$ (fig. S6).

Division	AA	AB	BB	N	p_{inv}	mean lat. ($^{\circ}\text{N}$)
6C	38	40	9	87	0.333	36.9
6B	34	47	16	97	0.407	38.3
6A	35	45	15	95	0.395	39.2
3N	0	0	1	1	1.000	43.1
4X	42	41	11	94	0.335	43.2
4W	21	17	9	47	0.372	43.6
3O	7	7	2	16	0.344	44.0
4T	7	11	5	23	0.457	47.7
4R	27	25	4	56	0.295	48.7
4S	8	4	3	15	0.333	49.9
3K	35	47	20	102	0.426	50.0
Total	254	284	95	633	0.374	—

Table S3: **Use of the two outgroups.** *I. coindetii* served two roles — ancestral-state polarization of the chromosome-2 inversion and the divergence term of the genome-wide McDonald–Kreitman (MK) test — both resting on a single whole-genome *I. coindetii*↔*Illex illecebrosus* AnchorWave alignment. *I. argentinus* was aligned identically (CDS) and used as a secondary MK outgroup; it yields $\sim 9\times$ fewer fixed coding differences than *I. coindetii* yet has genome-wide π indistinguishable from *Illex illecebrosus*, marking it as a recently-diverged congener of near-identical diversity.

Step	Quantity	Value
<i>Alignment</i> (AnchorWave v1.2.3, CDS-anchored)		
	<i>I. coindetii</i> substitution sites (genome-wide)	45,722,335
	aligned scaffold identity (inversion body)	99.2–99.9%
<i>Polarization</i> of chr2 karyotypes (est-sfs 2.04 + non-circular subset)		
	diagnostic (AA/BB fixed-difference) sites	13,998
	... where <i>I. coindetii</i> = <i>Illex illecebrosus</i> REF (circular; excluded)	9,680 (69.2%)
	... where <i>I. coindetii</i> carries the ALT (non-circular; used)	4,175
	polarity from non-circular set	BB derived / AA ancestral
<i>McDonald–Kreitman</i> test (mkado, <i>I. coindetii</i> divergence)		
	genes tested	20,999
	nonsynonymous / synonymous divergence (Dn / Ds)	15,343 / 13,670
	nonsynonymous / synonymous polymorphism (Pn / Ps)	468,124 / 630,949
	asymptotic α (95% CI)	0.642 (0.539–3.673)
<i>I. argentinus</i> (secondary MK outgroup; CDS alignment)		
	nonsynonymous / synonymous divergence (Dn / Ds)	1,355 / 1,805
	fixed coding differences (vs <i>I. coindetii</i> 29,013)	3,160 ($\sim 9\times$ fewer)
	asymptotic α	0.526
	genome-wide π (ANGSD, 6 autosomes; <i>Illex illecebrosus</i> = 0.0093)	0.0092

Table S4: **Gene-ontology enrichment of the Tier-1 (diploSHIC-only, Markov-corrected) sweep genes.** The 658 Tier-1 calls overlap 894 genes (476 GO-annotated); enrichment by Fisher’s exact test against the annotated-gene background on the scanned chromosomes, Benjamini–Hochberg FDR. 124 biological-process terms reach $FDR < 0.05$ — representative terms shown, spanning the two dominant themes (neural development; RNA-regulatory/transcriptional control). All are carried by the hard-sweep calls; the soft-sweep gene set has no term at $FDR < 0.05$. Full machine-readable table: `hmm_decode/tier1_GO_enrichment.tsv`.

GO id	term (biological process)	genes (T1/bg)	fold	FDR
GO:0048699	generation of neurons	124/1444	1.60	1.5×10^{-4}
GO:0022008	neurogenesis	126/1511	1.55	1.6×10^{-4}
GO:0030182	neuron differentiation	91/1076	1.57	1.5×10^{-3}
GO:0030154	cell differentiation	187/2553	1.36	3.7×10^{-4}
GO:0009887	animal organ morphogenesis	96/1116	1.60	8.3×10^{-4}
GO:0050767	regulation of neurogenesis	67/695	1.79	9.4×10^{-4}
GO:0000902	cell morphogenesis	72/776	1.72	1.3×10^{-3}
GO:0007409	axonogenesis	45/463	1.81	1.0×10^{-2}
GO:0048858	cell projection morphogenesis	59/610	1.80	1.6×10^{-3}
GO:0051252	regulation of RNA metabolic process	145/1922	1.40	1.3×10^{-3}
GO:2001141	regulation of RNA biosynthetic process	135/1770	1.42	1.5×10^{-3}

Table S5: **Functional clusters of the background-selection-robust Tier-1 sweep genes.** The 152 enriched biological-process GO terms ($FDR < 0.05$) of the BGS-robust set (fig. S15), grouped into functional clusters by term semantics. Neural development and cell morphogenesis/differentiation dominate, with RNA-regulatory control and organ development, and a distinct photoreception/sensory cluster.

functional cluster	terms	fold range	representative term (min FDR)
Neural development	22	1.6–22.2	generation of neurons (3×10^{-5})
Cell morphogenesis & differentiation	17	1.5–2.2	cell development (2×10^{-5})
RNA / transcription regulation	14	1.5–2.1	reg. RNA metabolic process (9×10^{-4})
Organ / anatomical development	13	1.2–2.5	system development (3×10^{-3})
Photoreception / sensory response	7	1.2–5.8	response to light stimulus (1×10^{-4})
Other regulation / metabolism	65	1.2–22.2	reg. developmental process

Table S6: **Tier-2 (cross-method-concordant) sweep genes.** The 34 genome-wide-scan regions where the diploSHIC-HMM sweep signal is independently corroborated by an LD/SFS-footprint method (RAiSD or SweepFinder2). Seven regions carry a named protein-coding gene (12 genes, listed); the remaining concordant regions are gene deserts or uncharacterised loci. Coordinates in Mb; α and DoS from the McDonald–Kreitman test against *I. coindetii* (‘–’ where the test is unpowered, i.e. too few fixed/polymorphic sites). The full candidate set — all 216 windows, the 34 concordant regions, and the per-region background-selection test — is provided as machine-readable tables (`outlier_scan_45/{windows,regions}.tsv` and the `concordant gene/MK/BGS` tables).

region (Mb)	sweep	gene	EnTAP best hit	α	DoS
1:20.3	hard	<i>Aplnr</i>	Apelin receptor	1.00	+0.17
1:85.5	hard	<i>eif3h</i>	Eukaryotic translation initiation factor 3	0.69	+0.20
1:85.5	hard	<i>babam1</i>	BRISC and BRCA1-A complex member 1	–	-0.48
1:85.5	hard	<i>ttc39b</i>	Tetratricopeptide repeat protein 39B	–	-0.27
5:38.0	hard	<i>cept1</i>	Choline/ethanolaminephosphotransferase 1	–	-1.00
5:38.0	hard	<i>Arsi</i>	Arylsulfatase I	–	-0.25
13:73.2	hard	<i>Rab3</i>	Ras-related protein Rab-3	–	-0.17
13:73.2	hard	<i>hyi</i>	Putative hydroxypyruvate isomerase	–	-0.64
13:73.2	hard	<i>Vps28</i>	Vacuolar protein sorting-associated protein 28	–	–
32:9.3	soft	<i>MacroD2</i>	ADP-ribose glycohydrolase MACROD2	–	-0.59
33:9.0	hard	<i>Tinagl1</i>	Tubulointerstitial nephritis antigen-like	–	-0.23
35:39.0	hard	<i>zag-1</i>	Zinc finger E-box-binding homeobox (ZEB)	–	-0.31

Table S7: **Genes captured by the chromosome-2 inversion body** (chr2:60.54–79.50 Mb). The 45 functionally annotated genes (of 89 gene models), ordered by position; *fixed* = number of exonic fixed differences between the AA and BB arrangements ($|\Delta AF| \geq 0.9$; ‘–’ none). EnTAP best-hit descriptions. The set spans lipid metabolism (phospholipases, PNPLA, malonyl-/acyl-CoA transferases), peroxisomal and mitochondrial function, DNA repair (XRCC1, TDG, WRNIP1), transcription/splicing (ERG, PRPF8, WDR82) and signalling (PRKCD), with no coherent pathway — consistent with the absent GO enrichment (fig. S21). Gene IDs are the LOC_ suffix.

region (Mb)	gene	EnTAP best hit	fixed
2:60.6	00005299	protein HGH1 homolog isoform X1	–
2:60.9	00005300	Phospholipase A2 domain-containing protein	2
2:61.4	00005301	lon protease homolog 2, peroxisomal isoform X2	–
2:61.4	00005302	39S ribosomal protein L18, mitochondrial	–
2:61.4	00005303	sodium- and chloride-dependent taurine transporter	–
2:61.6	00005304	cytochrome P450 4X1-like	4
2:61.6	00005305	uncharacterized protein LOC115219520	–
2:64.1	00005315	PNPLA domain-containing protein	–
2:64.1	00005316	PNPLA domain-containing protein	–
2:65.5	00005319	CCHC-type domain-containing protein	–
2:65.7	00005320	uncharacterized protein LOC127831186	–
2:66.1	00005322	solute carrier family 66 member 2 isoform X2	–
2:66.3	00005324	Ubiquitin-conjugating enzyme E2 T	–
2:68.0	00005330	PCDHD1 (protocadherin)	–
2:68.8	00005334	retrovirus-related Pol polyprotein (transposon)	–
2:69.0	00005337	PNPLA domain-containing protein	–
2:69.0	00005338	ELMO domain-containing protein	–
2:69.1	00005339	malonyl-CoA-acyl carrier protein transacylase, mito.	1
2:69.1	00005340	acidic repeat-containing protein-like	4
2:69.1	00005341	RPGRIP1L (ciliary)	2
2:70.2	00005347	transcriptional regulator ERG homolog	–
2:71.2	00005349	transcriptional regulator ERG homolog	–
2:71.6	00005350	uncharacterized protein LOC115215517	1
2:71.9	00005351	cysteine-rich venom protein	2
2:72.0	00005352	cysteine-rich venom protein	–
2:74.6	00005356	DUF7043 domain-containing protein	–
2:77.7	00005360	SWIM-type domain-containing protein	–
2:77.8	00005361	UPB1 (β -ureidopropionase)	–
2:77.8	00005362	DNA-repair protein XRCC1 (N-terminal domain)	–
2:77.8	00005363	TIM21 (mitochondrial import)	–
2:77.9	00005364	dynein light chain roadblock-type 2	–
2:77.9	00005365	TDG (thymine-DNA glycosylase)	–
2:78.0	00005367	small integral membrane protein 4	–
2:78.1	00005368	uncharacterized protein LOC115226124	–
2:78.1	00005369	ATPase WRNIP1-like	–
2:78.2	00005371	unnamed protein product (<i>Sepia</i>)	1
2:78.2	00005372	PPM1H (protein phosphatase)	–
2:78.2	00005373	WDR82	–
2:78.2	00005374	glycerate kinase-like	–
2:78.4	00005375	PRKCD (protein kinase C δ)	–
2:78.6	00005376	acyl-coenzyme A amino acid N-acyltransferase 1-like	–
2:78.7	00005378	uncharacterized protein LOC115226077	–
2:78.7	00005379	PRPF8 (pre-mRNA splicing factor)	–

region (Mb)	gene	EnTAP best hit	fixed
2:78.8	00005380	synaptic vesicle 2-related protein-like	–
2:79.3	00005384	unnamed protein product (<i>Sepia</i>)	–

Table S8: **Sampling events.** Every collection event (NAFO division, date, depth) with its source, the number of individuals analysed in this study (all 633 whole-genome sequenced individuals) and the number retained by [Baker et al. \(2025\)](#) after their genotype-missingness filter (540). Divisions are ordered north to south by mean sampling latitude. 4X was bulk-sampled across the division with no per-tow depth or position recorded.

Division	Date	Source	Depth (m)	<i>N</i> (this study)	<i>N</i> (Baker)
3K	16 August 2023	Recreational jig	3	48	48
	30 August 2023	Coastal fishery jig	5	54	46
4S	28 August 2023	Canadian trawl survey	228	2	2
	29 August 2023	Canadian trawl survey	251	1	1
	30 August 2023	Canadian trawl survey	80	1	1
	30 August 2023	Canadian trawl survey	124	2	2
	31 August 2023	Canadian trawl survey	152	2	1
	31 August 2023	Canadian trawl survey	155	3	3
	1 September 2023	Canadian trawl survey	273	1	1
	1 September 2023	Canadian trawl survey	296	1	1
	3 September 2023	Canadian trawl survey	243	2	2
	7 August 2023	Canadian trawl survey	113	36	24
4R	25 August 2023	Canadian trawl survey	173	2	2
	25 August 2023	Canadian trawl survey	233	5	4
	25 August 2023	Canadian trawl survey	245	9	7
	28 August 2023	Canadian trawl survey	221	4	3
	28 August 2023	Canadian trawl survey	221	4	3
4T	2 September 2023	Canadian trawl survey	243	1	0
	7 September 2023	Canadian trawl survey	27	1	1
	8 September 2023	Canadian trawl survey	49	1	0
	9 September 2023	Canadian trawl survey	112	1	0
	9 September 2023	Canadian trawl survey	183	1	0
	9 September 2023	Canadian trawl survey	306	1	0
	10 September 2023	Canadian trawl survey	81	1	1
	10 September 2023	Canadian trawl survey	101	1	1
	10 September 2023	Canadian trawl survey	285	1	0
	12 September 2023	Canadian trawl survey	285	1	0
	15 September 2023	Canadian trawl survey	50	1	1
	22 September 2023	Canadian trawl survey	39	1	1
	22 September 2023	Canadian trawl survey	142	2	1
	22 September 2023	Canadian trawl survey	158	1	1
	23 September 2023	Canadian trawl survey	341	2	2
	24 September 2023	Canadian trawl survey	118	1	0

Division	Date	Source	Depth (m)	N (this study)	N (Baker)
3O	27 September 2023	Canadian trawl survey	36	1	0
	28 September 2023	Canadian trawl survey	56	1	1
	28 September 2023	Canadian trawl survey	62	1	1
	29 September 2023	Canadian trawl survey	49	2	1
	13 October 2023	Canadian trawl survey	204	14	7
	14 October 2023	Canadian trawl survey	350	1	1
	14 October 2023	Canadian trawl survey	613	1	1
4W	16 July 2023	Canadian trawl survey	84	47	43
4X	July 2022	Canadian trawl survey	not recorded	94	76
3N	20 October 2023	Canadian trawl survey	160	1	1
6A	22 September 2022	US trawl survey	120	95	84
6B	18 June 2023	Commercial trawl	74	49	46
	16 August 2023	Commercial trawl	201	48	48
6C	14 September 2022	US trawl survey	307	87	73
Total				633	540

Table S9: **Elevated-diversity blocks.** Blocks of ≥ 1 Mb whose rolling nucleotide diversity exceeds twice the chromosome median. π fold: block mean over chromosome median. Genes per Mb from the annotation. Division ratio: range across the ten sampling divisions of block-to-flank diversity (5 Mb flanks, ANGSD).

Block	Mb	π	π fold	D block / chrom	Genes/Mb block / chrom	Division ratio
chr1:24.0–31.3	7.3	0.021	2.8	−1.77 / −2.21	1.8 / 6.1	2.20–2.37
chr6:12.0–18.9	6.9	0.018	2.3	−1.58 / −2.17	1.7 / 6.8	1.26–1.34
chr6:19.3–23.2	3.9	0.019	2.4	−1.56 / −2.17	1.8 / 6.8	1.27–1.32
chr8:29.7–35.5	5.8	0.018	2.3	−1.97 / −2.22	1.7 / 5.4	1.97–2.12
chr9:27.2–29.9	2.7	0.027	3.6	−1.12 / −2.12	2.2 / 6.5	3.01–3.19
chr10:66.7–69.4	2.7	0.018	2.1	−2.06 / −2.26	0.4 / 5.2	1.46–1.53
chr11:44.1–45.5	1.4	0.015	2.1	−1.83 / −2.15	2.1 / 8.4	1.48–1.53
chr13:33.5–39.0	5.5	0.024	2.7	−1.79 / −2.19	3.1 / 5.5	2.50–2.79
chr21:0.0–9.4	9.4	0.024	2.6	−1.39 / −1.94	1.7 / 6.4	2.22–2.49
chr21:74.4–79.4	5.0	0.023	2.5	−1.42 / −1.94	2.0 / 6.4	2.09–2.30
chr23:0.1–5.6	5.5	0.043	4.3	−0.96 / −2.04	2.2 / 5.8	3.94–4.43
chr23:62.4–65.3	2.9	0.023	2.3	−1.68 / −2.04	2.8 / 5.8	1.28–1.33
chr23:65.7–67.7	2.0	0.023	2.3	−1.73 / −2.04	1.5 / 5.8	1.23–1.27
chr25:61.4–63.8	2.4	0.023	2.8	−1.57 / −2.17	2.9 / 6.2	2.68–2.86
chr29:34.4–40.7	6.3	0.023	2.6	−1.41 / −1.97	2.1 / 7.9	2.44–2.79
chr30:17.3–18.5	1.2	0.057	6.1	−0.69 / −2.17	2.5 / 6.4	5.77–6.17
chr34:29.9–44.7	14.8	0.028	2.6	−1.45 / −1.95	1.8 / 5.2	2.20–2.43
chr35:59.7–65.1	5.4	0.033	3.4	−1.47 / −2.13	1.9 / 6.9	3.21–3.52
chr38:19.0–31.8	12.8	0.032	2.8	−1.41 / −1.97	2.2 / 5.2	2.70–3.08
chr38:61.5–68.0	6.5	0.048	4.1	−1.08 / −1.97	2.6 / 5.2	4.11–5.08
chr43:45.7–61.6	15.9	0.029	2.4	−1.52 / −1.84	2.2 / 6.8	1.83–2.02
chr44:8.8–11.7	2.9	0.028	2.7	−1.27 / −1.85	1.4 / 7.3	2.35–2.59

References

- Jeffrey R. Adrion, Jared G. Galloway, and Andrew D. Kern. Predicting the landscape of recombination using deep learning. *Molecular Biology and Evolution*, 37(6):1790–1808, 2020. doi: 10.1093/molbev/msaa038.
- Nikolaos Alachiotis and Pavlos Pavlidis. RAISeD detects positive selection based on multiple signatures of a selective sweep and SNP vectors. *Communications Biology*, 1:79, 2018. doi: 10.1038/s42003-018-0085-8.
- Krista D. Baker, Tony Kess, Lisa Hendrickson, Chelsea Malayny, Nicole Smith, and Ian R. Bradbury. Genomic data reveal a lack of range-wide genetic structure for a broadly distributed ommastrephid, Northern shortfin squid (*Illex illecebrosus*), in the Northwest Atlantic Ocean. *ICES Journal of Marine Science*, 82(11):fsaf201, 2025. doi: 10.1093/icesjms/fsaf201.
- Tobias Baril, James Galbraith, and Alex Hayward. Earl Grey: a fully automated user-friendly transposable element annotation and analysis pipeline. *Molecular Biology and Evolution*, 41(4):msae068, 2024. doi: 10.1093/molbev/msae068.
- Franz Baumdicker, Gertjan Bisschop, Daniel Goldstein, Graham Gower, Aaron P. Ragsdale, Georgia Tsambos, Sha Zhu, Bjarki Eldon, E. Castedo Ellerman, Jared G. Galloway, Ariella L. Gladstein, Gregor Gorjanc, Bing Guo, Ben Jeffery, Warren W. Kretzschmar, Konrad Lohse, Michael Matschiner, Dominic Nelson, Nathaniel S. Pope, Consuelo D. Quinto-Cortés, Murillo F. Rodrigues, Kumar Saunack, Thibaut Sellinger, Kevin Thornton, Hugo van Kemenade, Anthony W. Wohns, Yan Wong, Simon Gravel, Andrew D. Kern, Jere Koskela, Peter L. Ralph, and Jerome Kelleher. Efficient ancestry and mutation simulation with msprime 1.0. *Genetics*, 220(3):iyab229, 2022. doi: 10.1093/genetics/iyab229.
- Benjamin Buchfink, Klaus Reuter, and Hajk-Georg Drost. Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nature Methods*, 18(4):366–368, 2021. doi: 10.1038/s41592-021-01101-x.
- Petr Danecek, James K. Bonfield, Jennifer Liddle, John Marshall, Valeriu Ohan, Martin O. Pollard, Andrew Whitwham, Thomas Keane, Shane A. McCarthy, Robert M. Davies, and Heng Li. Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2):giab008, 2021. doi: 10.1093/gigascience/giab008.
- Sean R. Eddy. Accelerated profile HMM searches. *PLoS Computational Biology*, 7(10):e1002195, 2011. doi: 10.1371/journal.pcbi.1002195.
- Graham Gower, Aaron P. Ragsdale, Gertjan Bisschop, Ryan N. Gutenkunst, Matthew Hartfield, Ekaterina Noskova, Stephan Schiffels, Travis J. Struck, Jerome Kelleher, and Kevin R. Thornton. Demes: a standard format for demographic models. *Genetics*, 222(3):iyac131, 2022. doi: 10.1093/genetics/iyac131.
- gwct. degenotate: annotation of degeneracy of coding sites. <https://github.com/gwct/degenotate>, 2024. v1.3; no paper.
- Benjamin C. Haller, Peter L. Ralph, and Philipp W. Messer. SLiM 5: eco-evolutionary simulations across multiple chromosomes and full genomes. *Molecular Biology and Evolution*, 43(1):msaf313, 2026. doi: 10.1093/molbev/msaf313.

- Julien Jouganous, Will Long, Aaron P. Ragsdale, and Simon Gravel. Inferring the joint demographic history of multiple populations: beyond the diffusion approximation. *Genetics*, 206(3):1549–1567, 2017. doi: 10.1534/genetics.117.200493.
- Peter D. Keightley and Benjamin C. Jackson. Inferring the probability of the derived vs. the ancestral allelic state at a polymorphic site. *Genetics*, 209(3):897–906, 2018. doi: 10.1534/genetics.118.301120.
- Jerome Kelleher, Alison M. Etheridge, and Gilean McVean. Efficient coalescent simulation and genealogical analysis for large sample sizes. *PLoS Computational Biology*, 12(5):e1004842, 2016. doi: 10.1371/journal.pcbi.1004842.
- Jerome Kelleher, Kevin R. Thornton, Jaime Ashander, and Peter L. Ralph. Efficient pedigree recording for fast population genetics simulation. *PLoS Computational Biology*, 14(11):e1006581, 2018. doi: 10.1371/journal.pcbi.1006581.
- Andrew D. Kern and Daniel R. Schrider. discoal: flexible coalescent simulations with selection. *Bioinformatics*, 32(24):3839–3841, 2016. doi: 10.1093/bioinformatics/btw556.
- Andrew D. Kern and Daniel R. Schrider. diploS/HIC: an updated approach to classifying selective sweeps. *G3: Genes, Genomes, Genetics*, 8(6):1959–1970, 2018. doi: 10.1534/g3.118.200262.
- Thorfinn Sand Korneliussen, Anders Albrechtsen, and Rasmus Nielsen. ANGSD: analysis of next generation sequencing data. *BMC Bioinformatics*, 15:356, 2014. doi: 10.1186/s12859-014-0356-4.
- Sam Kovaka, Aleksey V. Zimin, Geo M. Pertea, Roham Razaghi, Steven L. Salzberg, and Mihaela Pertea. Transcriptome assembly from long-read RNA-seq alignments with StringTie2. *Genome Biology*, 20:278, 2019. doi: 10.1186/s13059-019-1910-1.
- kr-colab. pg_gpu: GPU-accelerated population genetics. https://github.com/kr-colab/pg_gpu, 2025. v0.1.1.dev14+g820c7da89; no paper.
- Heng Li. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics*, 34(18):3094–3100, 2018. doi: 10.1093/bioinformatics/bty191.
- Xiaoming Liu and Yun-Xin Fu. Stairway Plot 2: demographic history inference with folded SNP frequency spectra. *Genome Biology*, 21:280, 2020. doi: 10.1186/s13059-020-02196-9.
- Jaina Mistry, Sara Chuguransky, Lowri Williams, Matloob Qureshi, Gustavo A. Salazar, Erik L. L. Sonnhammer, Silvio C. E. Tosatto, Lisanna Paladin, Shriya Raj, Lorna J. Richardson, Robert D. Finn, and Alex Bateman. Pfam: the protein families database in 2021. *Nucleic Acids Research*, 49(D1):D412–D419, 2021. doi: 10.1093/nar/gkaa913.
- Francisco J. Pardo-Palacios, Angeles Arzalluz-Luque, Liudmyla Kondratova, Pedro Salguero, Jorge Mestre-Tomás, Rocío Amorín, Eva Estevan-Morió, Tianyuan Liu, Adalena Nanni, Lauren McIntyre, Elizabeth Tseng, and Ana Conesa. SQANTI3: curation of long-read transcriptomes for accurate identification of known and novel isoforms. *Nature Methods*, 21(5):793–797, 2024. doi: 10.1038/s41592-024-02229-2.
- Aaron R. Quinlan and Ira M. Hall. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics*, 26(6):841–842, 2010. doi: 10.1093/bioinformatics/btq033.

- Angel G. Rivera-Colón, Clara T. Rehmann, and Andrew D. Kern. MKado: a toolkit for McDonald–Kreitman tests of natural selection. *G3: Genes, Genomes, Genetics*, 16(8):jkag142, 2026. doi: 10.1093/g3journal/jkag142.
- Enrique Santiago, Carlos Köpke, and Armando Caballero. Accounting for population structure and data quality in demographic inference with linkage disequilibrium methods. *Nature Communications*, 16:6054, 2025. doi: 10.1038/s41467-025-61378-w.
- S. Small. msinv: ARG-based coalescent simulator with chromosomal inversions. <https://github.com/stsmall/msinv>, 2025. MIT license; commit abd0d63.
- Baoxing Song, Santiago Marco-Sola, Miquel Moreto, Lynn Johnson, Edward S. Buckler, and Michelle C. Stitzer. AnchorWave: sensitive alignment of genomes with high sequence diversity, extensive structural polymorphism, and whole-genome duplication. *Proceedings of the National Academy of Sciences*, 119(1):e2113075119, 2022. doi: 10.1073/pnas.2113075119.